ENTERPRISE SYSTEM ARCHITECTURE

IMA CAMPUS MASTER SYSTEM SPECIFICATION

A comprehensive technical manual, departmental workflow guide, and complete service directory detailing all capabilities, modules, and operational logic.

MULTI-TENANT CORE

Unified enterprise engine managing multiple institutes, batches, courses, faculty, and cross-business logistics.

LUMI AI INTELLIGENCE

Autonomous voice agent, automated receipt verification OCR, live analytics, and generative marketing studios.

4-TIER CAMPAIGN HUB

Advanced lifecycle CRM managing Open Seminar, Inbound Inquiries, Bridge Transfers, and Retention funnels.

SUPPLY CHAIN LOGISTICS

End-to-end tute warehouse management, hold resolution workflows, and integrated courier tracking pipelines.

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> **Document Scope & Confidentiality Notice**
>
> This specification serves as the definitive reference document for all technical workflows, data models, user interface states, and operational procedures implemented within the IMA Campus web and mobile ecosystems.

01 System Architecture & Multi-Tenant Hierarchy

IMA Campus operates on an enterprise-grade **Multi-Tenant Relational Architecture** built to scale across multiple educational institutes, diverse academic disciplines, and hundreds of thousands of concurrent students.

1.1 Data Model Entity Relationship

Entity Level	Database Schema Model	Operational Purpose & Functionality
1. Business (Tenant)	`Business` / `CrmSettings`	Represents an autonomous educational brand/institute. Owns isolated Meta WhatsApp Cloud API credentials, Gemini AI API keys, payment merchant IDs, custom call ringtones, and default medium rules.
2. Batch (Cohort)	`Batch` / `BatchMonthlyStats`	Represents an academic year or target examination cohort (e.g., 2026 A/L, 2027 A/L). Sets coordinator KPI benchmarks: Individual Goal (40%) and Group Goal (50%).
3. Group (Stream)	`Group`	Academic subject streams (e.g., Physical Science, Biological Science, Commerce, Technology, Arts). Houses bundle discount matrices and installment rules.
4. Course (Subject)	`Course` / `TuteStock`	Individual subjects mapped to specific lecturers, monthly fee schedules, bundle exclusions, active academic lifespans, and physical warehouse inventory.

02 User Roles & Role-Based Access Control (RBAC)

The system enforces strict zero-trust Role-Based Access Control (RBAC), providing customized dashboards, dedicated navigation bars, and granular data access according to user roles and departmental affiliations.

User Role	Department Mapping	Permissions & System Capabilities
SYSTEM_ADMIN / SUPER	Executive Administration	Unrestricted full-system access: CRM configuration, database manager, test injector, staff profile modifications, payment overrides, and **Ghost Mode**.
HEAD MANAGER	Executive Operations	Multi-branch oversight, staff KPI appraisals, social media delegation approval, and executive finance matrix inspection.
CC MANAGER	Class Coordination	Coordinator performance monitoring, quota distribution, lecturer tute calendar verification, and roster approval.
COORDINATOR	Class Coordination	Free Seminar CRM, 4-tier Call Campaigns, Class Timetable, Task Center, Monthly Roster, Web Inquiries, and Student Data Center.
CALLER	Calling Unit	Lead execution dialer, disposition recording, feedback entry, and call attempts logging.
CALL CENTER MGR	Call Center	Agent workload balancing, real-time call volume analytics, break time tracking, and bridge transfer approvals.
CALL CENTER AGENT	Call Center	Dedicated call center dialer workspace, agent status switcher (Online, Lunch, Away), and fast lead escalation.
SOCIAL MEDIA MGR	Social Media	Marketing job card review, task allocation to designers/editors, content calendar planning, and AI creative tools.
SOCIAL MEDIA STAFF	Social Media	Assigned graphic/video design execution, submitting proof assets, version revisions, and AI subtitle video editing.
DELIVERY STAFF	Delivery & Logistics	Order packaging, hold issue resolution, waybill generation, courier dispatch, and warehouse tute stock tracking.
FINANCE STAFF	Finance & Accounts	Manual & AI receipt verification, post-pay scheduling, bank statement reconciliation, and payment dispute handling.
PHYSICAL MGR	Physical Branches	RFID card issuance, smart student attendance, lecturer revenue shares, hall scheduling, and branch petty cash ledgers.
TECHNICAL STAFF	Technical Operations	Studio hall allocation, Zoom API credentials rotation, secure anti-piracy watermarked player streaming, and incident logs.
STUDENT	Public Student Portal	V2 Dashboard, live watermarked video player, delivery tracking timeline, Paycorp gateway, and moderated app community chat.

03 Class Coordination Department

The Class Coordination Department serves as the operational backbone of IMA Campus. It bridges student inquiries, faculty scheduling, tute delivery schedules, daily task tracking, and lead conversions.

3.1 Core Modules & Daily Coordinator Workflows

1. Coordinator Daily Hub & KPI Matrix

Live trackers displaying the coordinator's daily target progress against Individual Goals (40%) and Group Goals (50%), remaining daily calling quotas, pending task tallies, and active classes scheduled for the day.

2. Class Timetable Engine

Full scheduling interface for live classes with start/end times, assigned hall, streaming links (Zoom, YouTube, Vimeo), assigned coordinator, and automated generation of pre/in/post class checklists.

3. Template-Driven Task Center

Custom and automated recurring tasks with deadline enforcement. Coordinators complete duties by uploading screenshot proof. Overdue tasks are locked and require manager approval with written justification to unlock.

4. Lecturer Tute Stock Calendar

Compares scheduled class dates against current warehouse inventory. Automatically warns coordinators when physical stock falls below enrolled student counts prior to a lecture.

5. Monthly Staff Roster

Structured duty schedule covering Normal Duty, Half Day, Day Off, and Poya Day shifts. Integrates daily manual call logs to measure individual team productivity.

6. Web Inquiries (LUMI Live Chat)

Real-time omnichannel student support interface. Inquiries from the website and portal are auto-routed to coordinators with instant chat, quick replies, and file attachments.

04 Free Seminar CRM & Lead Ingestion

The Free Seminar CRM captures, qualifies, and nurtures prospective students who register for free promotional seminars, converting them into active, fee-paying enrollments.

4.1 Autonomous Ingestion & Lead Distribution

- **Meta Cloud API Integration:** Direct webhook synchronization across multiple WhatsApp Business numbers. Messages instantly create leads with complete conversational metadata.
- **Bulk CSV/Excel Ingestion:** Ingests external marketing campaign data with automatic phone number normalization and deduplication safeguards.
- **Auto Assign Quota Engine:** Dynamically distributes incoming leads among on-duty coordinators using fair round-robin allocation bounded by configurable daily manager quotas.

4.2 WebRTC WhatsApp Voice Calling & Headless IVR

Coordinators initiate direct WhatsApp voice calls to prospective students straight from the browser using custom SIP/WebRTC bridges.

Interactive Permission & AI Audio Processing

Before outbound calls are triggered, an interactive WhatsApp template with Accept/Decline action buttons is dispatched. Upon call termination, the recording is transcribed into English and Sinhala, followed by Gemini AI sentiment analysis to output Mood, Summary, and Vibe Scores.

05 After Seminar CRM & The 4 Call Campaigns

Following a seminar, leads transition into the **4-Tier Call Campaign Suite**, an advanced pipeline designed to drive course enrollments, prevent student dropouts, and manage overdue fee collections.

5.1 Comprehensive Breakdown of the 4 Campaign Tiers

Campaign Tier	Target Audience & Ingestion Source	Operational Logic & Progression Rules
1. Open Seminar Leads	Students who registered for or attended free promotional seminars.	Multi-round calling (R1 → R2 → R3). Tracks calling attempts, contact statuses (Answered, Busy, Switched Off, Thinking), and auto-calculates delay penalties (`r1DelayedDays`) if follow-ups exceed SLA limits.
2. New Inquiries	Direct inbound hot leads from web forms, WhatsApp, and digital ads.	Enforces a high-urgency SLA countdown timer. Requires immediate first-call resolution and scheduling of follow-up callbacks.
3. Bridge Transfers	Leads transferred from Call Center agents or cross-business brands.	Displays complete previous conversation notes and transfer reasons, facilitating seamless context-aware re-engagement.
4. Paid Students (Retention)	Enrolled students with installment plans or dropped payment history.	Tracks installment schedules (Installments 1, 2, 3), post-pay grace periods, recovers dropped students, and drives upgrades from single subjects to full bundles.

5.2 Campaign Audit Trail & Status State Machine

Every lead status update, call attempt, and coordinator note is permanently immutably logged into `CallCampaignAuditLog`, ensuring complete transparency and accountability across teams.

06 Call Center Department

The Call Center Department acts as a centralized high-velocity calling engine, filtering massive volumes of raw leads and escalating qualified prospective students to class coordinators.

Call Center Manager Hub

- Live monitoring of inbound and outbound call traffic.
- Real-time agent state oversight (Active, In-Call, Lunch, Away).
- Dynamic lead pool rebalancing and bulk reassignment.
- Approval pipeline for bridge transfer requests to coordinators.

Call Center Agent Workspace

- High-speed one-click softphone dialer.
- Immediate disposition selection with standardized notes.
- Instant escalation: Single-click Bridge Handover to Class Coordination.
- Agent performance scorecard tracking daily completed calls.

07 Social Media Department & AI Creative Studio

The Social Media Department centralizes marketing requests, creative asset production, content calendar scheduling, and AI-powered multimedia generation across all campus brands.

7.1 Social Media Delegation Hub (Digital Job Cards)

Class coordinators and branch managers request promotional banners, video flyers, and reel scripts through structured **Digital Job Cards** with strict milestone auditing.

```
[Job Requested] ──> [Assigned] ──> [In Progress] ──> [Proof Submitted] ──> [Approved / Completed]
                                                                  |
                                                                  └──> [Correction Requested] ──> [Revised]
```

7.2 AI Creative Studio Tool Suite

Tool Name	Technology Engine	Capabilities & Output Deliverables
AI Video Generator	Generative Scripting + AI TTS	Generates complete promotional shorts/reels with synchronized voiceovers, background footage, and kinetic text from a text prompt.
AI Reel Music Matcher	Audio Vibe & Trend Classifier	Analyzes video theme and pacing to recommend top trending audio tracks for TikTok and Instagram algorithms.
AI Subtitle Video Editor	Speech Recognition + Auto-SRT	Transcribes video dialogue and burns in stylized, animated, high-engagement subtitles (CapCut-style formatting).
LUMI Image Generator	High-Resolution Image Models	Creates custom marketing graphics, subject artwork, and promotional backdrops on demand.

08 Delivery Department (Warehousing & Logistics)

The Delivery Department manages physical printed tute distribution from payment approval through courier waybill generation to final doorstep delivery.

8.1 Delivery Hub Pipeline Tabs

Dashboard Tab	Functional Operations
1. Overview	Executive summary metrics showing orders packed today, pending dispatches, active delivery holds, and completed shipments.
2. Pending & Holds	Isolates orders stalled by incorrect phone numbers, ambiguous addresses, or stock shortages. Features a coordinator collaboration loop to fix and auto-release held orders.
3. Delivered / Dispatched	Tracks shipped parcels with live courier tracking numbers, bulk waybill printing, and delivery date stamping.
4. Delivery History	Complete search archive with instant indexing by student name, telephone number, payment ID, or courier tracking code.
5. Tute Stock & Reports	Live warehouse inventory balance, minimum threshold alerts, auto-deduction upon packing, and historical audit logs.

09 Finance Department & Payment Management

The Finance Department oversees student billing, payment verification, installment schedules, anti-fraud slip checks, and bank statement reconciliations.

9.1 LUMI AI Slip Verification & Anti-Fraud Engine

Uploaded payment slips are immediately analyzed by Gemini Vision AI to prevent revenue leakage and automate approval workflows:

OCR & Metadata Extraction

Extracts deposited amount, transaction date/time, branch code, and reference numbers with confidence scoring.

Anti-Fraud & Tamper Detection

Flags digitally manipulated (Photoshop) receipts, font inconsistencies, and altered numbers.

Duplicate Receipt Detection

Prevents cross-student slip re-use by matching image hashes and reference numbers against historical submissions.

Bank Statement Reconciliation

Matches uploaded bank statement CSV feeds against approved student claims, verifying actual funds received.

9.2 Ledger Balancing & Installment Pricing

- **Excess & Arrears Management:** Overpayments or underpayments are automatically credited or debited to the student's digital ledger balance.
- **Upgrade Pricing Matrix:** Automatically calculates exact pro-rated price differentials when students upgrade from single subjects to multi-subject combo bundles.

10 Physical Institute CRM (Campus Operations)

Manages on-premises physical campus branches, walk-in student registrations, RFID smart attendance, faculty revenue sharing, and counter cash registers.

Module	Key Features & Capabilities
RFID Registration	Issues smart RFID cards linked to student profile, photo, school, grade, and guardian contact numbers.
Smart Attendance	Single-tap card scanning at branch turnstiles with automated real-time SMS/WhatsApp entry alerts dispatched to parents.
Faculty Management	Configurable institute commission percentages (Institute Cut %), hall utilization logs, and teacher payout payroll summaries.
Cashflow & Counter Register	Real-time tracking of on-site registration fees, daily petty cash disbursements, utility costs, and daily cash reconciliation.

11 Technical Department & Live Streaming

Coordinates live broadcast infrastructure, studio allocations, Zoom account credential rotation, and secure anti-piracy video delivery.

11.1 Anti-Piracy Watermarked Video Player

Live streams and recorded classes are rendered through an encrypted video player with a dynamic floating watermarking layer. The student's full name, telephone number, and IP address animate randomly across the screen, deterring and identifying unauthorized screen recording.

11.2 Zoom Multi-Account Provisioning

Maintains a pool of Zoom OAuth API accounts, automatically provisioning live class meetings, managing concurrent attendee limits, and injecting live access links directly into student dashboards.

12 LUMI AI Suite (Autonomous AI Engines)

The LUMI AI Suite integrates enterprise artificial intelligence across all core business operations:

LUMI Analyser
Predictive analytics forecasting cohort drop-off rates, future monthly revenue, and coordinator KPI benchmarks.

LUMI Chat
Smart NLP support bot providing instant, contextual answers to student inquiries 24/7.

LUMI Caller
Autonomous conversational voice agent conducting seminar reminder calls in natural Sinhala/English.

LUMI Image Studio
Generative marketing tool crafting high-resolution promotional banners and graphic assets.

13 Student Portal & Mobile App Ecosystem

The public-facing student environment is optimized for seamless learning, secure video streaming, and parcel tracking.

- **Student Dashboard V2:** Next-generation interface highlighting today's scheduled classes, live lecture join links, countdown timers, and campus announcements.
- **Delivery Tracking Hub:** Live multi-stage parcel progress tracker (Order Approved → Packed → Dispatched → Out for Delivery).
- **Paycorp Payment Gateway:** Multi-channel digital fee collection supporting Credit/Debit Cards, QR payments, and manual slip uploads.
- **Moderated Student Community Chat:** Dedicated batch-level discussions featuring automated AI content moderation that removes inappropriate language in real time.

14 System Administration & Core Utilities

Ghost Mode (Security Impersonation)

Super Admins can log into any coordinator, manager, or staff account without a password for instant troubleshooting. An active red banner allows one-click return to the admin session.

System Issue & Bug Tracker

In-app reporting system allowing staff to log software glitches with screenshot attachments directly to system developers.

Global Refresh Engine

Instant cache clearing and real-time data synchronization triggerable across all client sessions simultaneously.

Personalization & HUD

Theme engine supporting Dark/Light modes, dynamic animated backgrounds (Live Sky, Color Shifts), and an embedded Spotify player for staff focus.